



Python Programming - 2301CS404

Lab - 4

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01) WAP to print 1 to 10.

```
In [7]: for i in range(1,11,1):  
        print(i)
```

1
2
3
4
5
6
7
8
9
10

02) WAP to print 1 to n.

```
In [9]: number =int(input("Enter The Number To print From 1 To Number: "))  
        for i in range(1,number+1,1):  
            print(i)
```

1
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18

03) WAP to print odd numbers between 1 to n.

```
In [12]: num = int(input("Enter The Number To Print Odd From 1 To N: "))  
         for i in range(1,num+1,2):  
             print(i)
```

1
3
5
7
9
11
13
15
17
19

04) WAP to print numbers between two given numbers which is divisible by 2 but not divisible by 3.

```
In [35]: start = int(input("Enter The Starting Number Of Range: "))  
         end = int(input("Enter The Ending Number Of Range: "))  
  
         for i in range(start, end+1, 1):  
             if (i % 2 == 0) and (i % 3 != 0):  
                 print(i)
```

2
4
8
10
14
16

05) WAP to print sum of 1 to n numbers.

```
In [40]: numbers = int(input("Enter The Number To Print The Sum Of Numbers: "))  
         sumNumber = 0  
         for i in range(1,numbers+1,1):
```

```
sumNumber += i;
print(sumNumber)
```

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06) WAP to print sum of series $1 + 4 + 9 + 16 + 25 + 36 + \dots n$.

```
In [53]: nums = int(input("Enter The Number To Print The Square Of Numbers: "))
squareNumber = 0
for i in range(1, nums+1, 1):
    square = i**2
    squareNumber += square
print(squareNumber)
```

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07) WAP to print sum of series $1 - 2 + 3 - 4 + 5 - 6 + 7 \dots n$.

```
In [63]: series = int(input("Enter The Number of Odd Even Addition Subtraction: "))
sums = 0
for i in range(1, series+1, 1):
    if (series % 2 == 1):
        sums += i
    else:
        sums -= i
print(sums)
```

-3

08) WAP to print Multiplication Table of the given number.

```
In [46]: tableNumber = int(input("Enter The Number TO Print Multiplication Table: "))
for i in range(1, 11, 1):
    print(f"{tableNumber} * {i} = {tableNumber * i}")
```

```
5 * 1 = 5
5 * 2 = 10
5 * 3 = 15
5 * 4 = 20
5 * 5 = 25
5 * 6 = 30
5 * 7 = 35
5 * 8 = 40
5 * 9 = 45
5 * 10 = 50
```

09) WAP to find Factorial of the given number.

```
In [54]: FactNumber = int(input("Enter The Number To Find Factorial: "))
FactNum = 1
for i in range(1, FactNumber+1, 1):
    FactNum *= i
print(FactNum)
```

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10) WAP to print GCD of given two numbers.

```
In [68]: num1 = int(input("Enter The First Number To Find GCD: "))
num2 = int(input("Enter The Second Number To Find GCD: "))
gcd = 1
for i in range(num1,num2+1,1):
    if(num1 % i == 0) and (num2 % i == 0):
        gcd = i
print(f"gcd of {num1} and {num2} = {gcd}")
```

gcd of 5 and 15 = 5

11) WAP to find Factors of the given number.

```
In [72]: FactorNumber = int(input("Enter The Number To Find Factorial: "))
print(f"Factor of {FactorNumber} Are:")
for i in range(1, FactorNumber+1 , 1):
    if(FactorNumber % i == 0):
        print(i)
```

Factor of 8 Are:

1
2
4
8

12) WAP to find whether the given number is Prime or not.

```
In [5]: prime = int(input("Enter The Number To Check Number Is Prime Or Not: "))
count = 1
for i in range(2,prime+1,1):
    if (n % i == 0):
        count + 1
        break;

if (count == 2):
    print(f"{prime} is Prime Number")
else:
    print(f"{prime} is Not a Prime Number")
```

3 is Not a Prime Number

13) WAP to print sum of digits of given number.

```
In [107... digit = input("Enter The Number To Print Sum of Digit")
sumNumbers = 0
digitlength = len(digit)
digit = int(digit)
for i in range(1,digitLength+1,1):
    remainder = digit % 10
    sumNumbers += remainder
    digit = digit // 10

print(sumNumbers)
```

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14) WAP to check whether the given number is Palindrome or not.

```
In [110]: palindromeNum = input("Enter a number")
if palindromeNum == palindromeNum[::-1]:
    print("Yes its a palindrome")
else:
    print("No, its not a palindrome")
```

Yes its a palindrome

15) WAP to check whether the given number is an Armstrong Number or not.

```
In [2]: armstronNum = int(input("Enter a number: "))
n = armstronNum
power = len(str(armstronNum))
total = 0

while n > 0:
    digit = n % 10
    total += digit ** power
    n //= 10

if total == armstronNum:
    print("Armstrong Number")
else:
    print("Not an Armstrong Number")
```

Armstrong Number

16) WAP to print all the perfect numbers between 1 to n.

```
In [1]: NumInput = int(input("Enter the upper limit (n): "))

print(f"Perfect numbers between 1 and {NumInput} are:")

for num in range(1, NumInput + 1):
    divisorsSum = 0
    for i in range(1, num):
        if num % i == 0:
            divisorsSum += i

    if divisorsSum == num and num != 0:
        print(num)
```

Perfect numbers between 1 and 50 are:

6

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